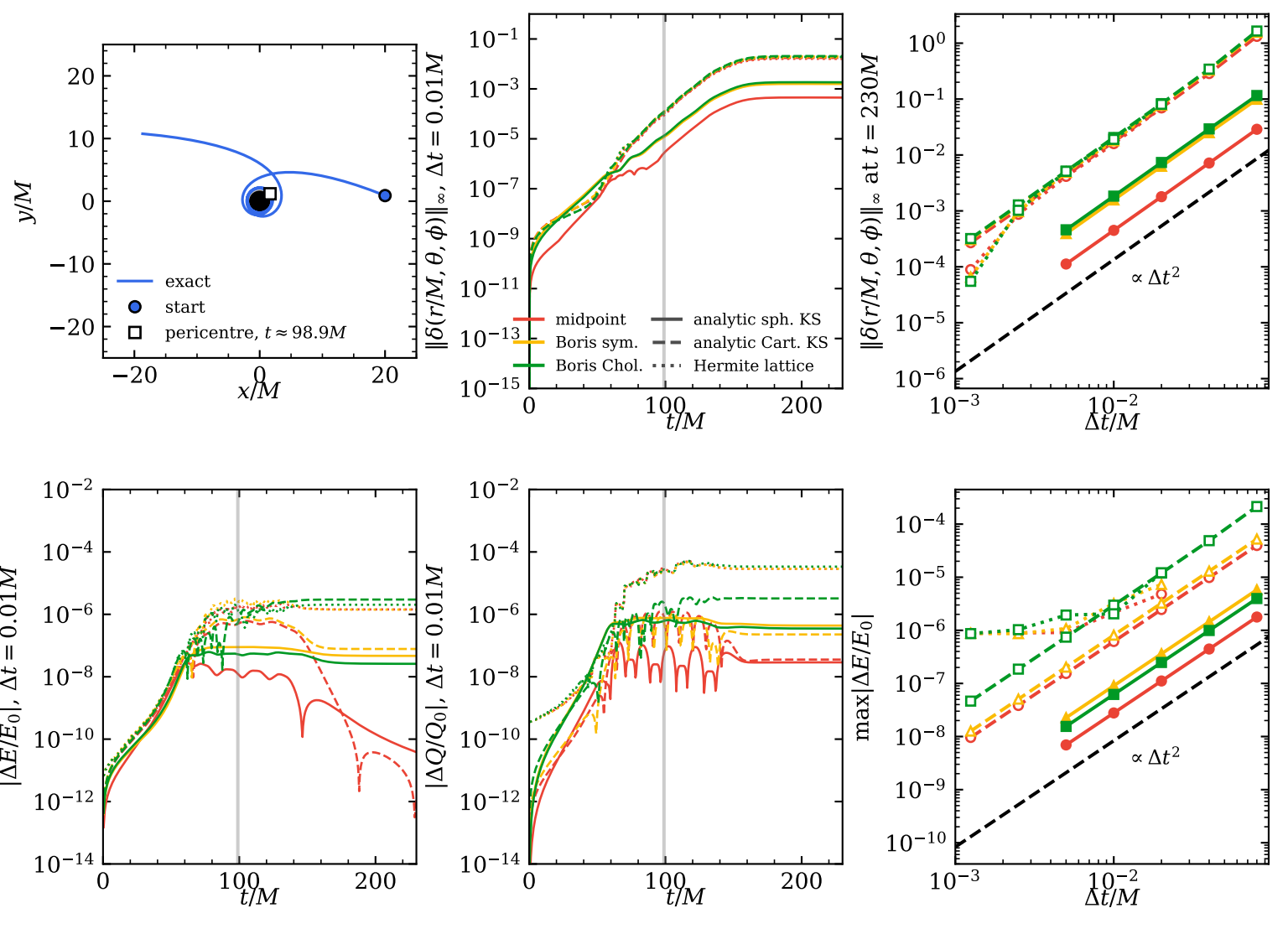

y/M
x/M
exact
start
pericentre, $t \approx 98.9M$
$\|\delta(r/M, \theta, \phi)\|_\infty$, $\Delta t = 0.01M$
t/M
midpoint
Boris sym.
Boris Chol.
analytic sph. KS
analytic Cart. KS
Hermite lattice
$\|\delta(r/M, \theta, \phi)\|_\infty$ at $t = 230M$
$\propto \Delta t^2$
$\Delta t/M$
$|\Delta E/E_0|$, $\Delta t = 0.01M$
$|\Delta Q/Q_0|$, $\Delta t = 0.01M$
$\max|\Delta E/E_0|$